

The background is a vibrant, low-poly 3D game environment. A large, glowing red sun dominates the upper left, casting a warm, reddish light over the scene. The terrain is composed of various low-poly shapes, including rocky hills and a flat, reddish ground. In the foreground, a character with a colorful, multi-colored body is riding a skateboard, leaving a motion blur trail behind them. The overall aesthetic is reminiscent of indie game art styles like those found in 'Cuphead' or 'Hollow Knight'.

Game Studies & Game Design

Josef Who?

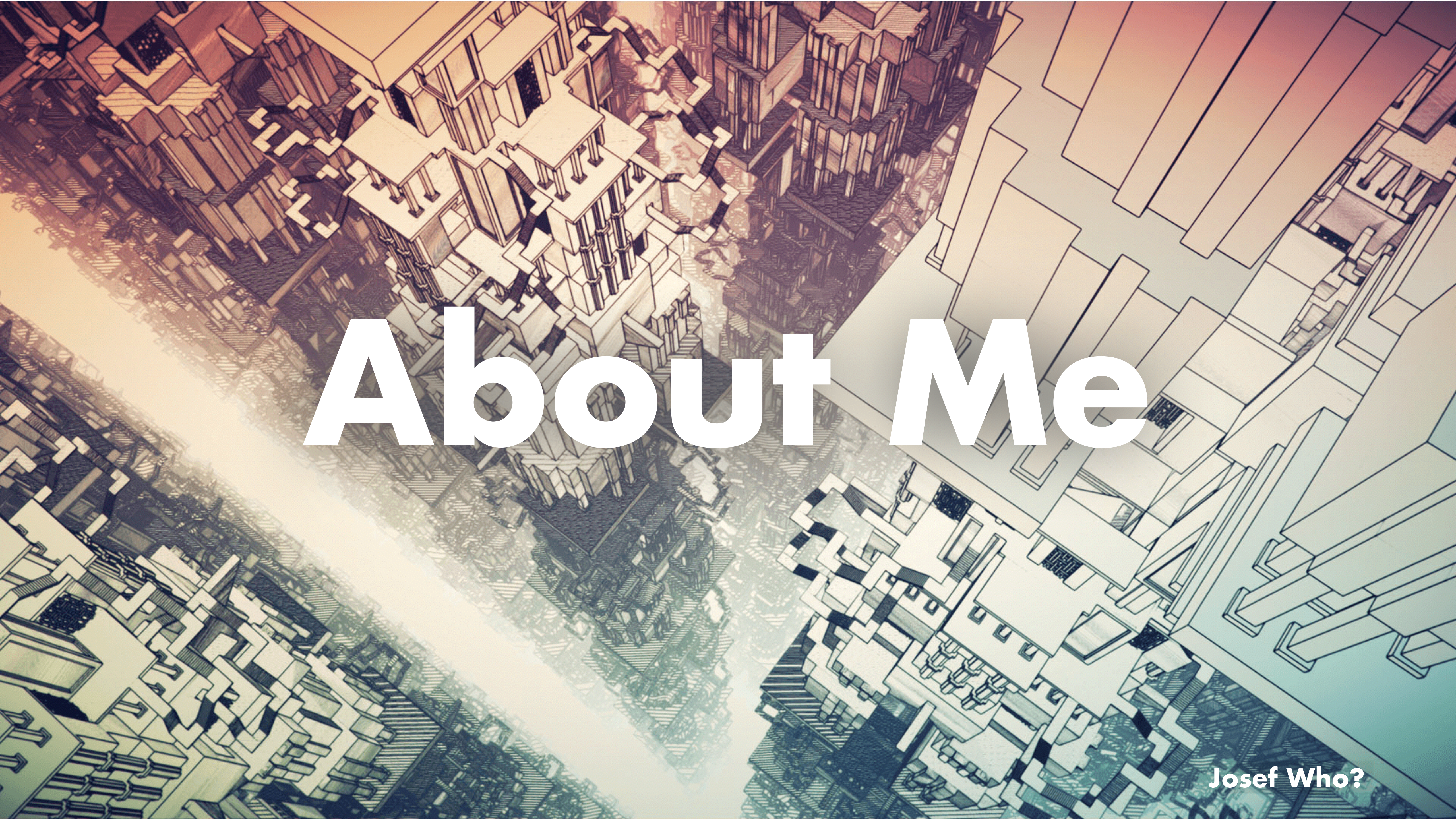


Do you know the game?



Skate Story

Sam Eng



About Me

Josef Who?



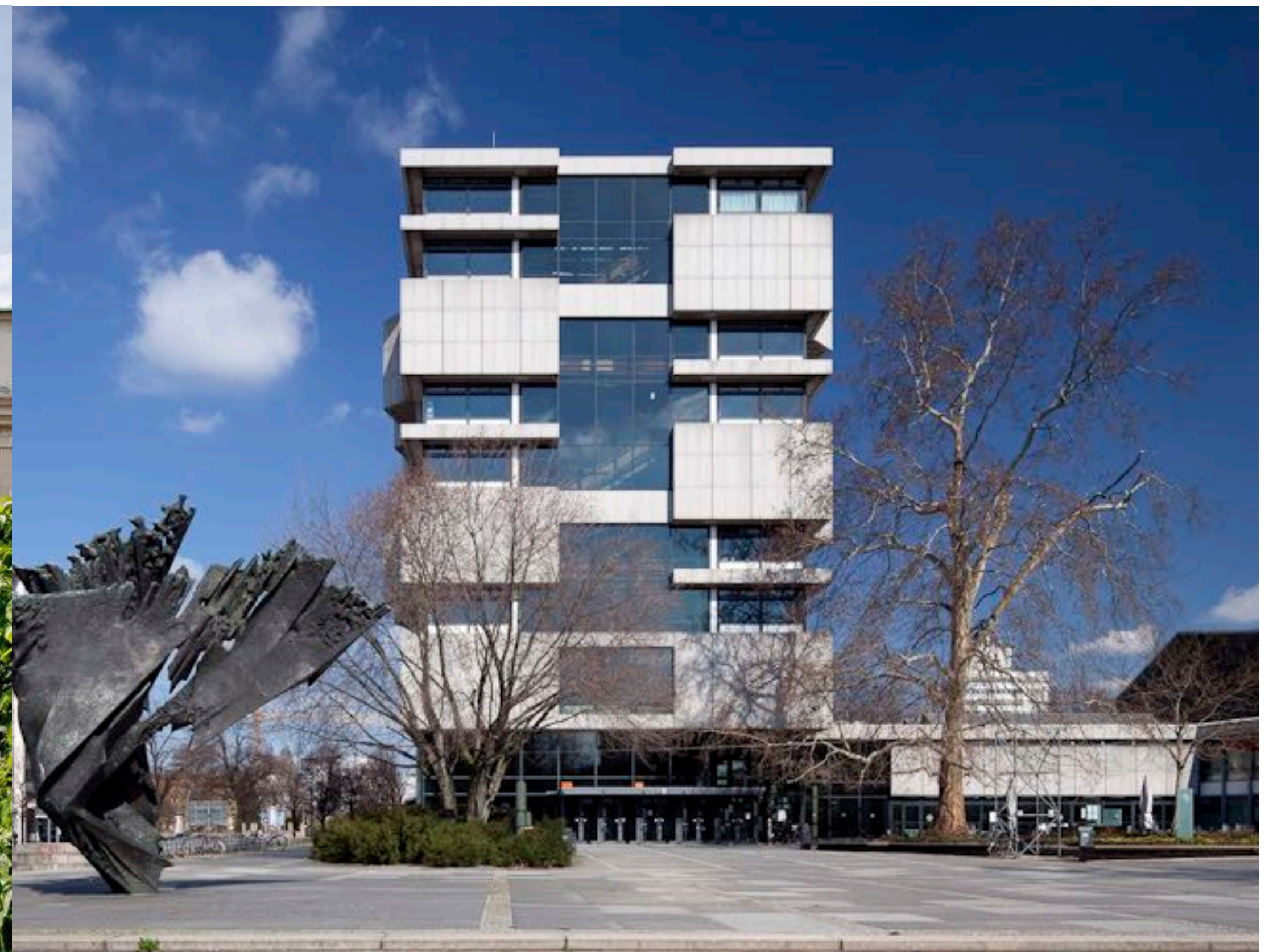
Do you know the game?



Manifold Garden

William Chyr

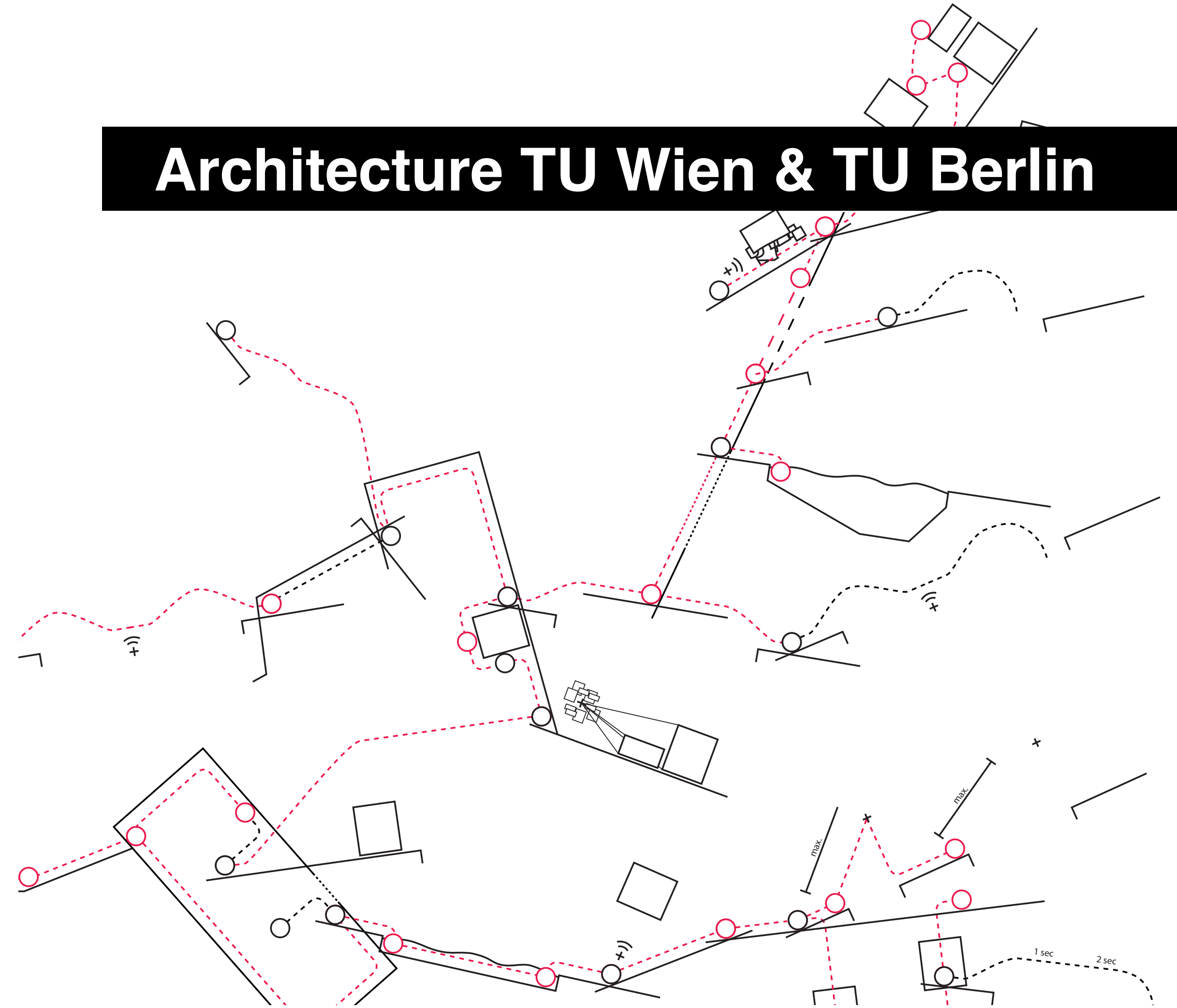
Architecture TU Wien & TU Berlin



Josef Who?

Räumlichkeit in digitalen Spielen

Untersuchung zu Strukturen und
Detaillösungen in virtuellen Spielräumen

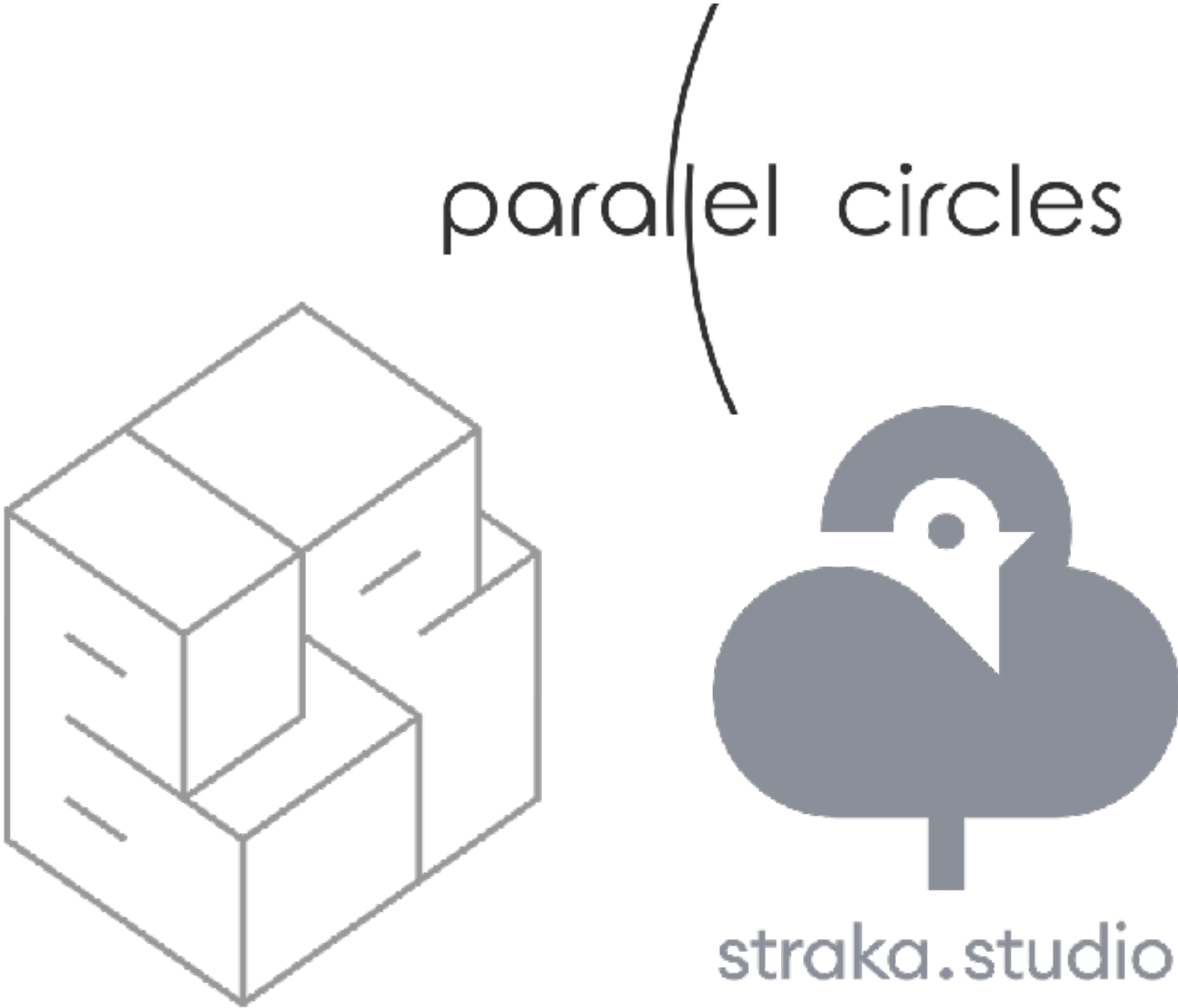




Josef Who?

since 2009

&





o-cim and yet it moves

And Yet It Moves

Broken Rules

PC, WiiWare, iOS

Level Design & Porting

Josef Who?



Chasing Aurora

Broken Rules

WiiU launch title

Level Design & Porting



Josef Who?

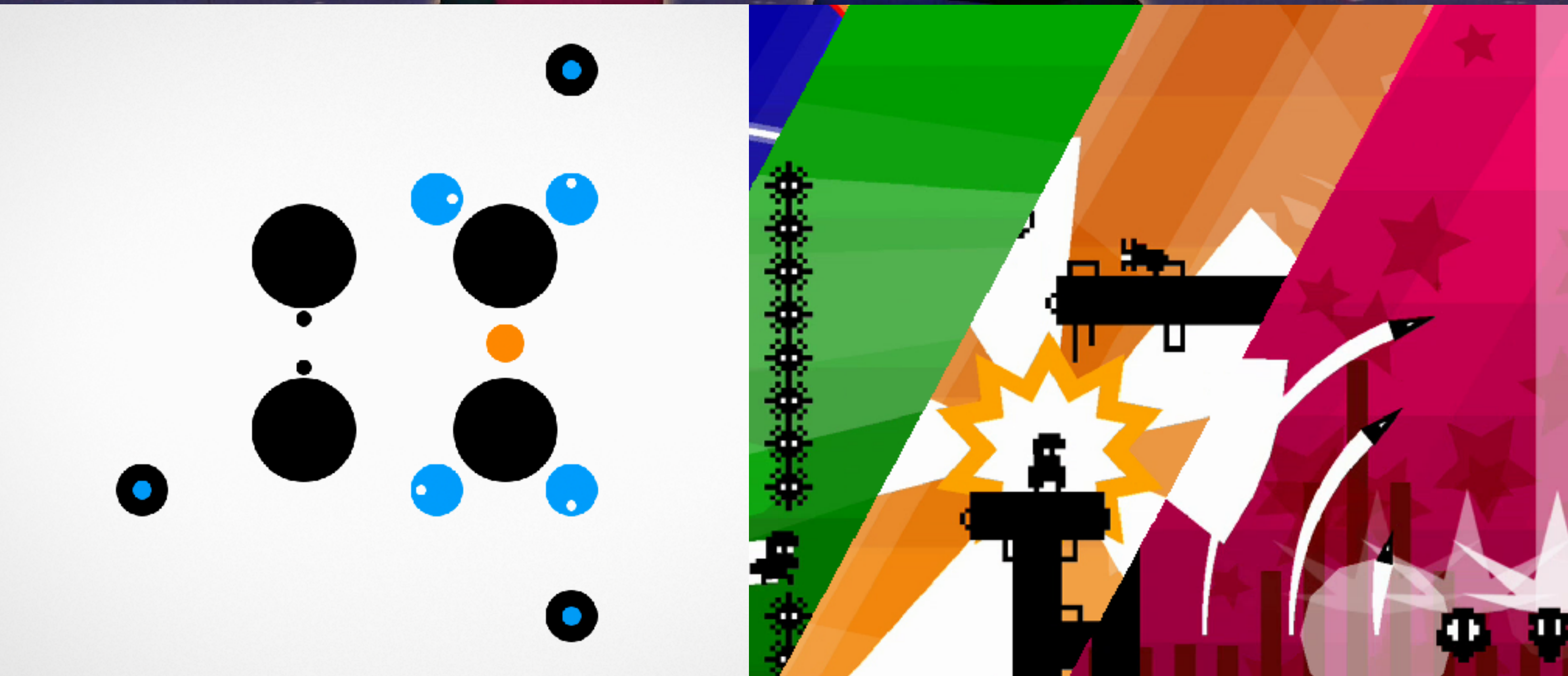


Guacamelee!

Blek

Electronic Super Joy

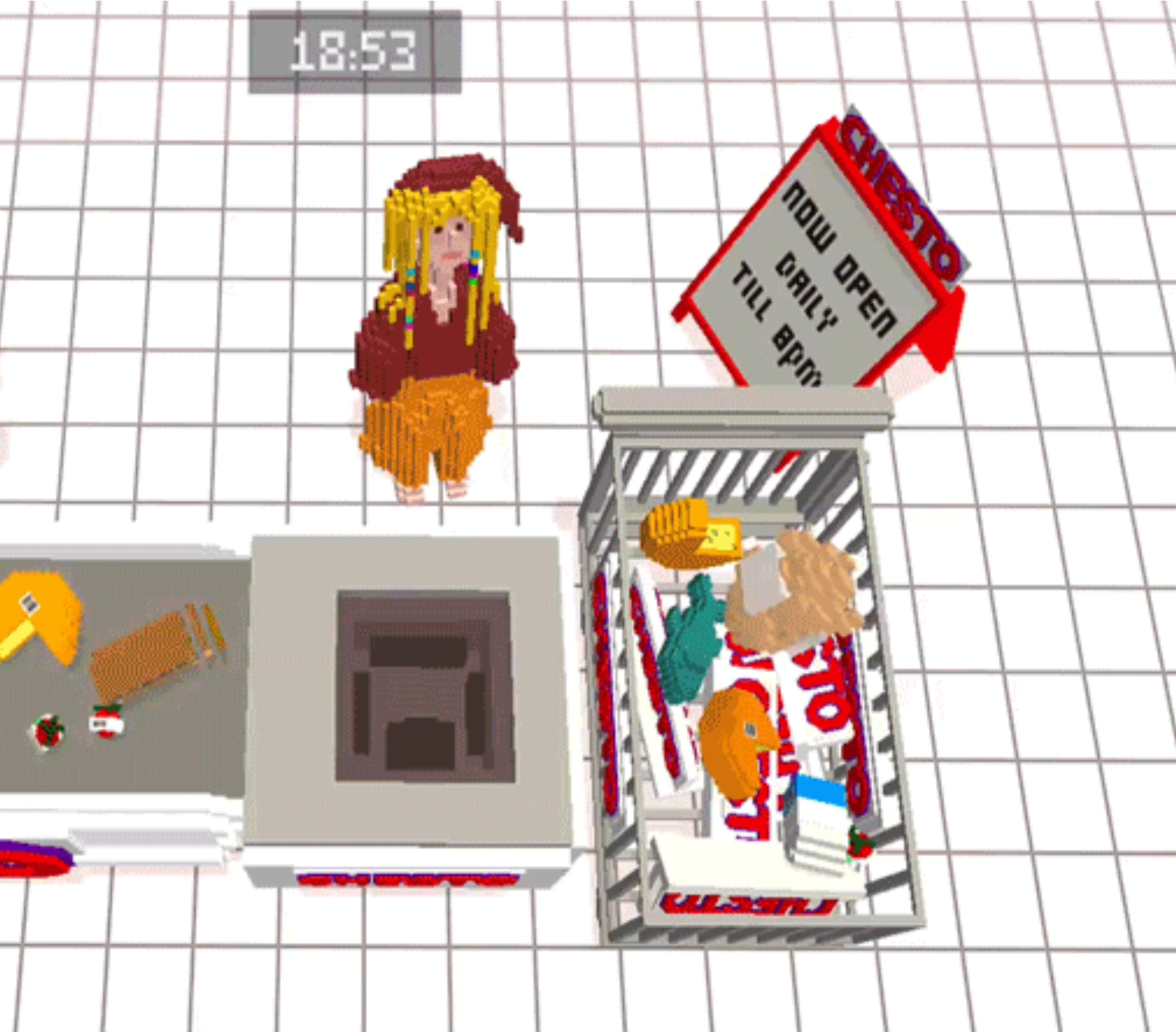
Broken Rules



WiiU porting



Josef Who?



CHESTO - At the Checkout

#altgame

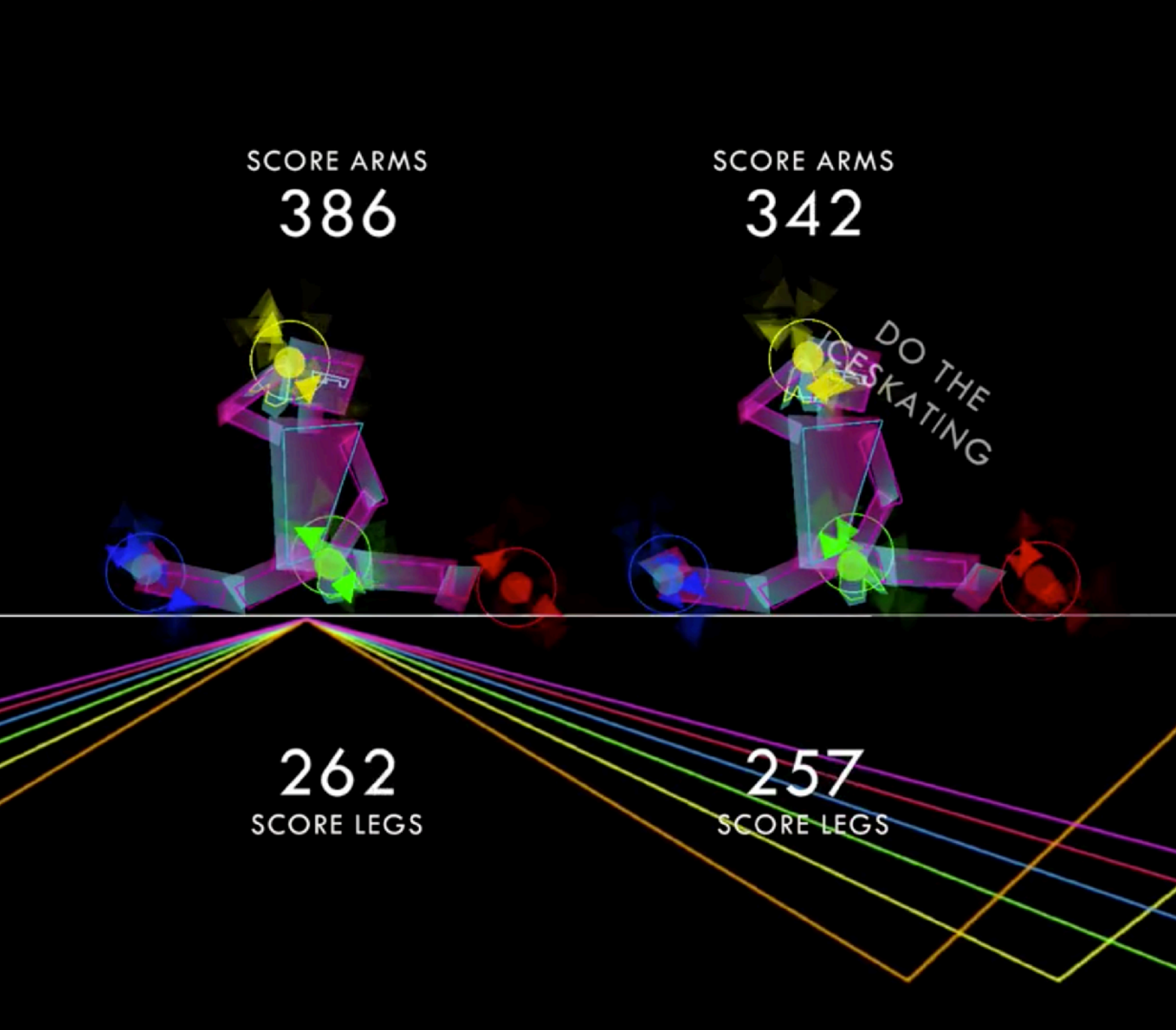
itch.io

Broken Rules

Game Design, Art,
Level Design



Josef Who?



Voguing Robotics

Shared Avatar

Voguing & Posing

Broken Rules

**Game Design, Art,
Level Design, Code**

Josef Who?



Parley Who Vertigo

Non-screen Game

Move Controller

Minigame Collection

Tischtennisringerl

Thomas Perl

**Game Design,
Environment Design**



Josef Who?



vinylOS

alt.ctrl

Turntable

Audio signal

Jonas Bohatsch

**Game Design, Code,
Art, Exhibition Design**



Josef Who?



Old Man's Journey

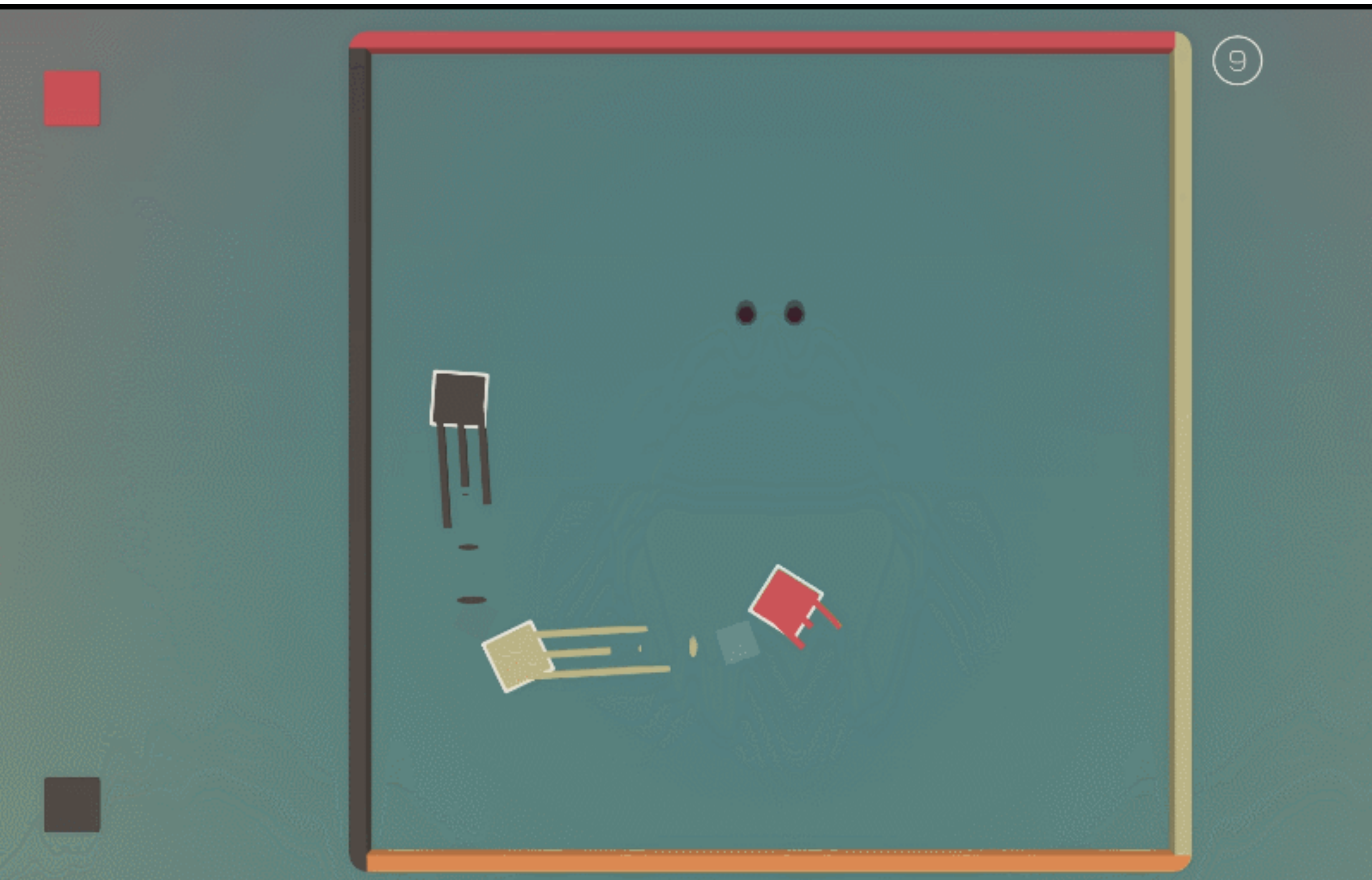
Switch, PS4, Xbox

Broken Rules

**Console porting, QA,
Publishing,
Store Management**



Josef Who?



Flat Heroes

Switch, PS4, Xbox, PC

Parallel Circles

**Console porting,
Publishing,
Store Management**

Josef Who?



Loot River

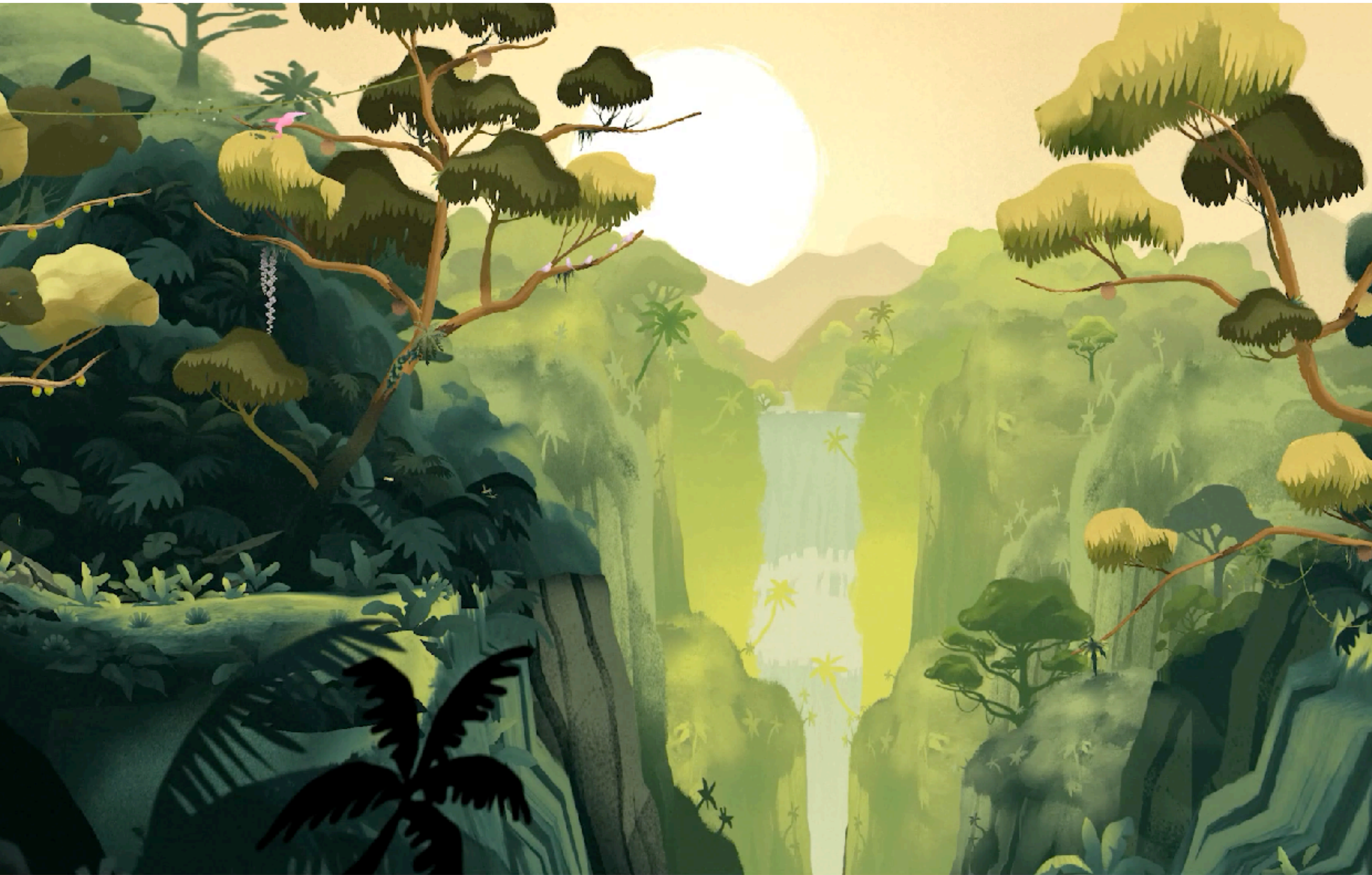
Xbox

straka.studio

Xbox publishing



Josef Who?



Gibbon: Beyond the Trees

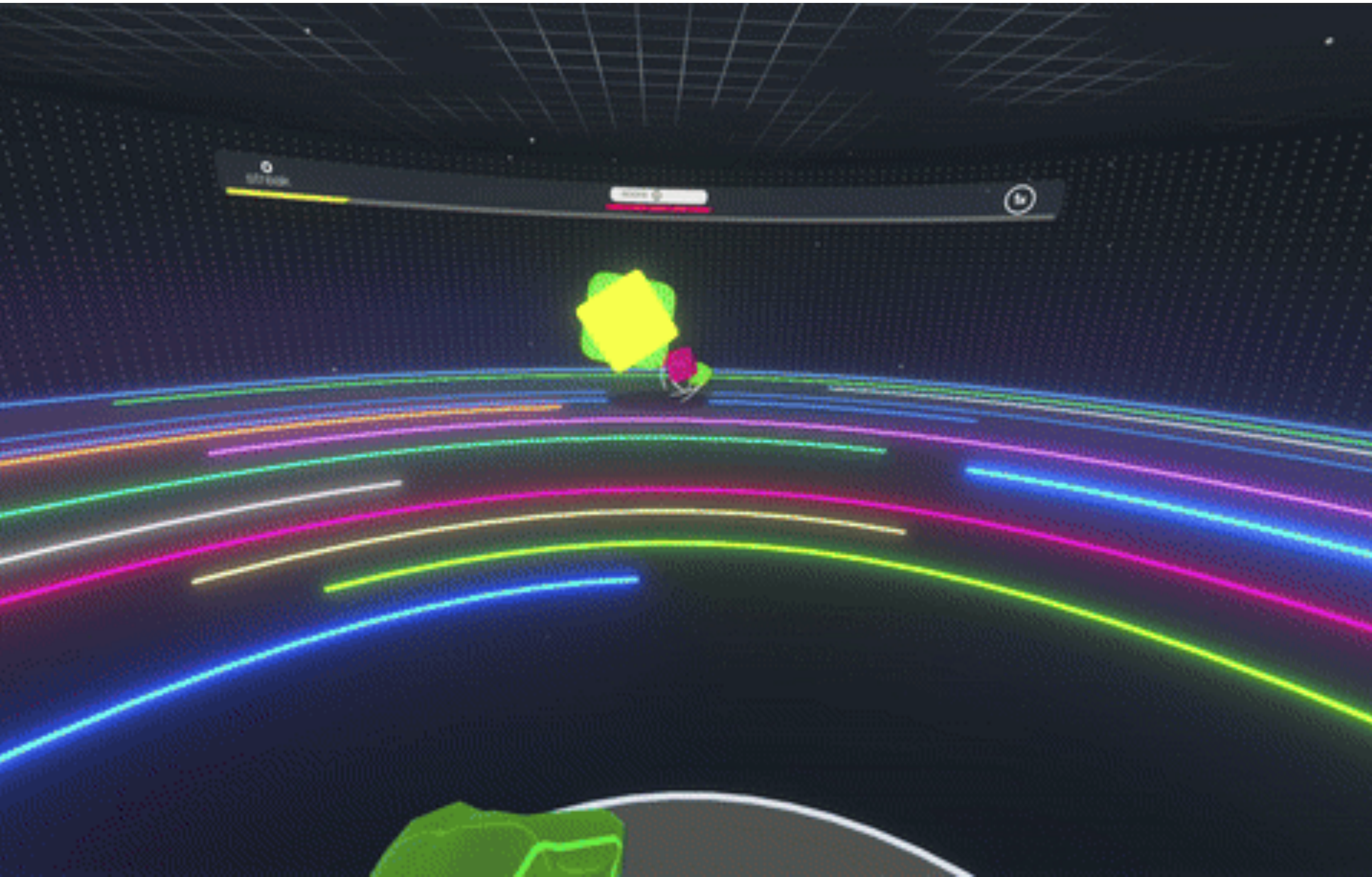
**Switch, PS4, PS5, PC,
Apple Arcade**

Broken Rules

**Level Design, Game
Design, Production,
Porting, Store
Management**



Josef Who?



Beat the Beats VR

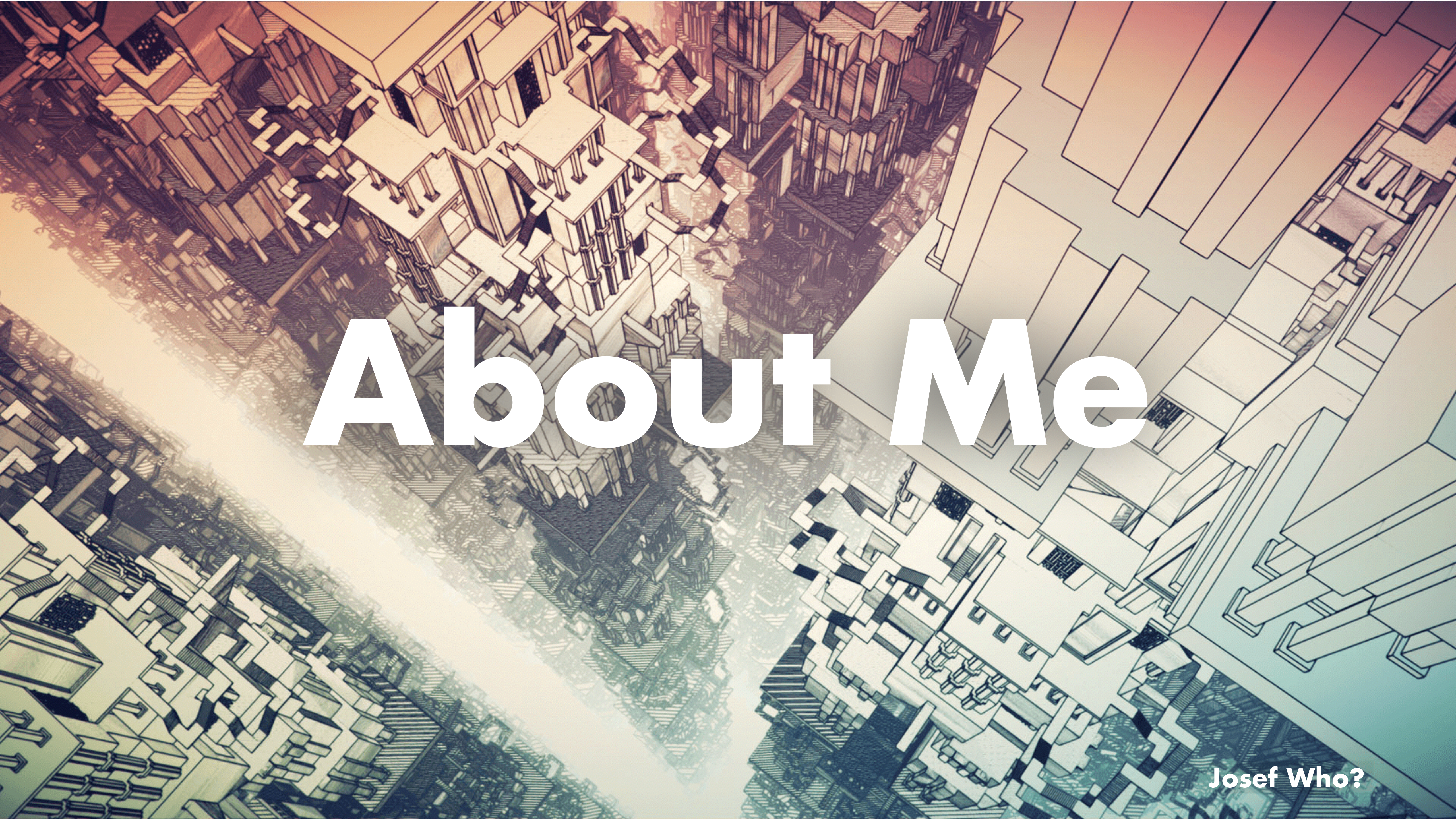
PlayStation VR2

Parallel Circles

**Publishing, QA,
Store Management**



Josef Who?

An aerial, top-down view of a dense urban landscape. The buildings are rendered in a stylized, blocky manner. A diagonal color gradient is applied across the image, transitioning from a warm orange-red at the top left to a cool teal-blue at the bottom right. The text 'About Me' is centered horizontally and vertically in a large, white, sans-serif font.

About Me

Josef Who?

A cartoon illustration of a boy and a dog. The boy, on the left, has a mustache, wears a brown hat, a yellow shirt, and blue overalls. He is standing on a green lawn next to a brown dirt mound. The dog, on the right, is white with large black spots and is lying on a red and white striped picnic blanket. The dog's head is turned towards the boy, and its eyes are wide open. The background features green grass, small green bushes, and a few small white flowers with yellow centers.

vo overview

Josef Who?



Do you know the game?



Thank Goodness You're Here

Coal Supper

Game Studies & Game Design 1, WS 2025

What? Different parts of game design.

Game Studies & Game Design 2, SS 2026

How? Processes of game design.

VO Structure

4 blocks in WS 2025

7x 45 minutes units -> approx 5 hours

1 presentation

Today

Nov

Nov

Jan

Josef Who?

VO Structure

4 blocks in WS 2025

7x 45 minutes units -> approx 5 hours

1 presentation

Theory

ME <—> YOU

Today

Nov

Nov

Jan

Josef Who?

VO Structure

4 blocks in WS 2025

7x 45 minutes units -> approx 5 hours

1 presentation

Theory

ME <—> YOU

Hands-On

Mini Workshops

Analysis

Today

Nov

Nov

Jan

Josef Who?

Goals

Understanding and use of:
Game Mechanics, Game Rules,
Game Systems, Interactivity,
Progression, Balancing,
Emergence, Probability,
Randomness, Gameplay,
Gameplay Loops, Player
Experience, Player Psychology,
...

Today

Nov

Nov

Jan

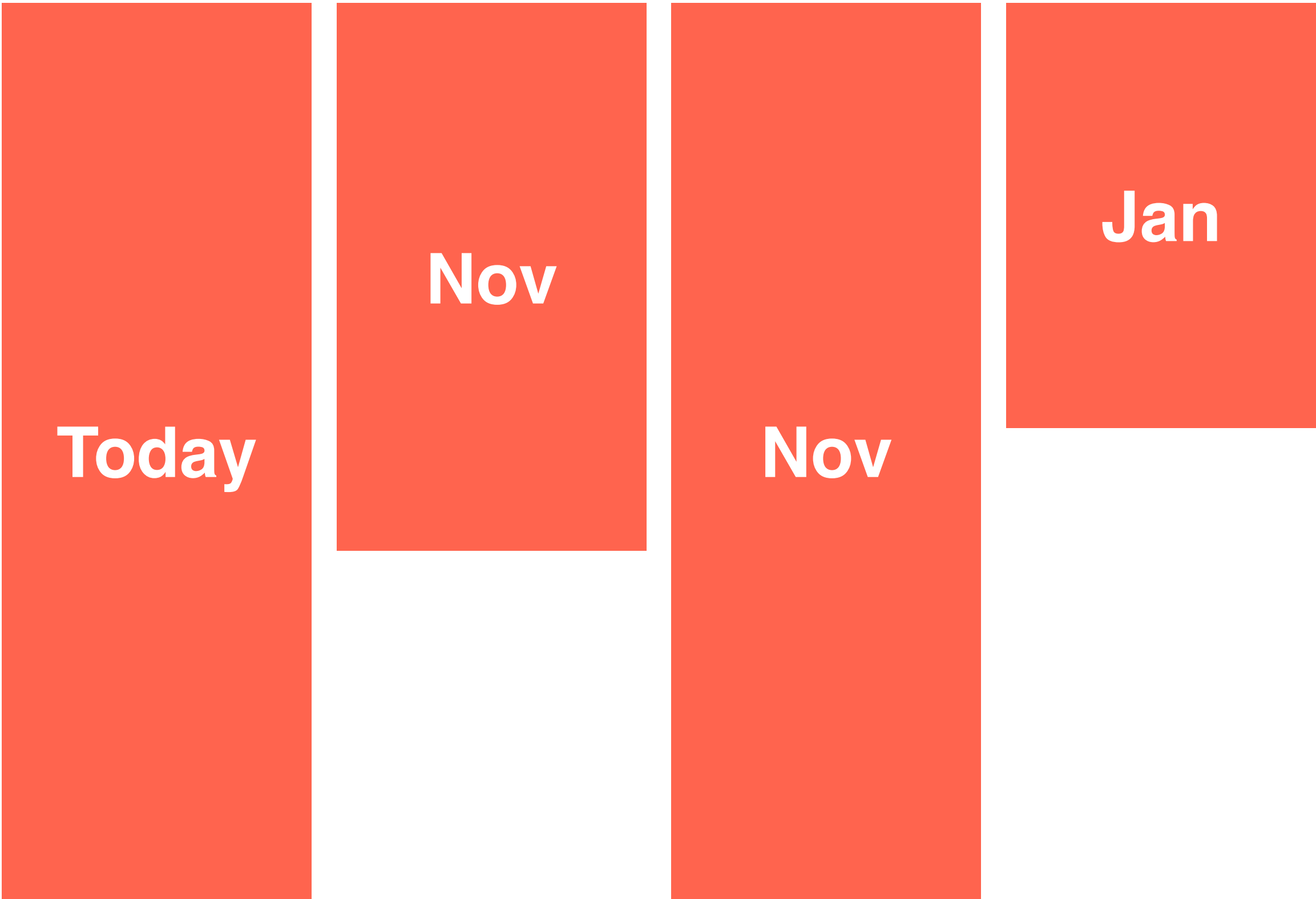
VO Structure

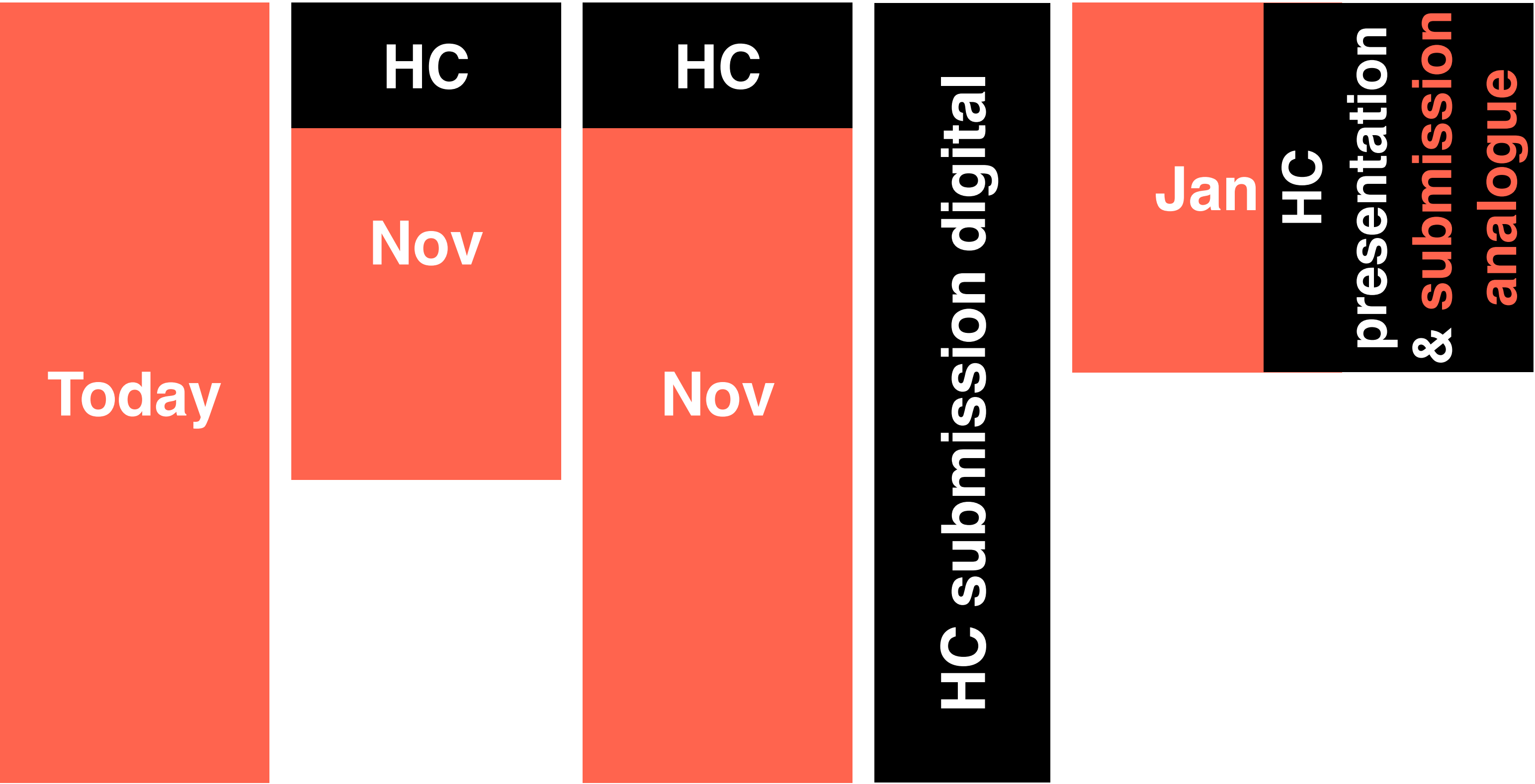
Grading

15% Attendance

15% Participation

70% High Concept





Topics

Definition

Game Mechanics: Rules

Interactivity

Hands-On

Today

- Topics
- Definition
- Game Mechanics: Rules
- Interactivity
- Hands-On

09:00 - 12:30 Game Studies & Game Design 1 (MMAB3WG1IL) Lehrveranstaltung 4 EH mma-b3 Game A, mmt-b3 GameDesign A WS Wiesner, Josef
13:30 - 16:00 Game Studies & Game Design 1 (MMAB3WG1IL) Lehrveranstaltung 3 EH mma-b3 Game A, mmt-b3 GameDesign A WS

Today
Day Structure?

Attendance

A cartoon illustration of a boy and a dog. The boy, on the left, has a mustache, wears a brown hat, a yellow shirt, and blue overalls. He is standing on a green lawn next to a brown dirt mound. The dog, on the right, is white with large black spots and is lying on a red and white striped picnic blanket. The dog's head is turned towards the boy, and its eyes are wide open. The background features green grass, small green bushes, and a few small white flowers with yellow centers.

vo overview

Josef Who?

Game Design?

Josef Who?



Do you know the game?



Dungeons of Hinterberg

Microbird Games

What is Game Design?

Goals

Game Design?

Who is a game designer?

What do I do as a game designer?

How do I do game design?

What is Game Design?

Definition

What is Game Design?

What do you 🦊 think?

Josef Who?

Definition

Design?

Eric Zimmerman

“Design is the process of learning how to solve problems.”



Josef Who?

What is Game Design?

Robert Zubek, Elements of Game Design

3 elements

- **Game mechanics and systems:** rules and objects of the game
- **Gameplay:** interaction between the player and the mechanics and systems of the game
- **Player Experience:** how players feel when playing the game

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Robert Zubek, Elements of Game Design

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Addition Wikipedia: “Game design is part of a game's development from concept to final form. Typically, the development process is iterative, with repeated phases of testing and revision...”

What is Game Design?

Daniel Klein, What is Game Design, medium.com

“...every single definition of Game Design I’ve found on the Internet is bad, actively harmful, misleading...”

“Game Design is a discipline involved in game development. Game Design is primarily concerned with player psychology: how will players feel when they experience this? What will they be thinking about? What will they expect, what will they want to do?”

What is Game Design?

Daniel Klein, What is Game Design, medium.com

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play 3 elements

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Daniel Klein, What is Game Design, medium.com

Game Design vs. Game Development

Anyone who works on a game is a game developer.

BUT

E.g. an animator is a game developer but not per se a game designer. Even if part of what they do can be described as game design.

Daniel Klein, What is Game Design, medium.com

Game Design vs. Game Development

Distinction pretty academic and abstract.

Not very helpful for someone who wants to get into game design.

Let's try to find out what a Game Designer is by defining what they actually do! And how they do it.

Daniel Klein, What is Game Design, medium.com

Game Design: What?

A game designer does not require an computer at all to engage in game design!

Most of the time a game designer works within systems.

As a game designer it is your job to understand these systems, what they are achieving so you can design whatever you are making to fit them.

Daniel Klein, What is Game Design, medium.com

Game Design: What?

Most of what a game designer creates can be described as mechanics.



Josef Who?

Daniel Klein, What is Game Design, medium.com

Game Design: What?

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Robert Zubek, Elements of Game Design

3 elements

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Daniel Klein, What is Game Design, medium.com

Game Design: What?

Most of what a game designer creates can be described as mechanics.

They will mostly be variations or new combinations of what already exists.

With your decision-making you are highly influencing game feel.

Daniel Klein, What is Game Design, medium.com

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Robert Zubek, Elements of Game Design

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Josef Who?

Daniel Klein, What is Game Design, medium.com

Game Design: What?

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Finally you look into everything that affects balance and you tune it accordingly so every made decision fits into the larger systems that exist.

Game Design: What?

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3 elements

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Daniel Klein, What is Game Design, medium.com

Game Design: How?

Ideating or finding and selecting what you want to build

**What is already there?
Why is it there?
How does it interact?**

Brainstorming together

Paperkitting

Prototyping (can still be analogue) or kit bashing

-> Playtesting

Collaborative and iterative!



Josef Who?

Daniel Klein, What is Game Design, medium.com

Game Design: How?

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How does it interact?

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IS IT FUN?

Josef Who?

Daniel Klein, What is Game Design, medium.com

Game Design: How?

Playtest your design to get feedback

Analyse feedback and iterate

Playtest iteration

Analyse feedback

...

Josef Who?

Daniel Klein, What is Game Design, medium.com

Game Design: How?

Production

Make your design real

Scripting and working with other tools

Working closely with other departments

You are in charge of the feeling and vision you are building

It helps if you as the game designer have some basic knowledge of the work of other departments



Josef Who?

Game Design?

A stylized illustration of a character in a red hood and green shirt standing in a field, holding a sword and a box labeled 'SUPER CUT 20'. In the background, there are mountains, a cable car, and a red tower.

Josef Who?

Rules

Josef Who?



Do you know the game?



Johann Sebastian Joust

Die Gute Fabrik

Rules are one of the essential qualities of games

Every game has a set of rules

Every set of rules defines a game

Goals

What are rules?

How do they function?

What is their relevance for game design?

What are game rules?

Example: Tic Tac Toe Gruppe A

What do you 🦊 think are the rules of the game?

Let's try to be as concise as possible!

- 2 players; turn based game
- each players has a symbol
- 3 by 3 grid
- setting 1 symbol per turn
- win if 3 in the same line
- only one symbol in one grid space
- all grid filled and no one wins it is a draw
- (you must set a symbol)

O	O	O
	O	X
	X	X

Josef Who?

What are game rules?

Example: Tic Tac Toe Gruppe B

What do you 🦊 think are the rules of the game?

Let's try to be as concise as possible!

- turn based for 2 players
- you have to place different symbols
- 3 by 3 grid
- 3 in a row to win
- place one symbol at a time ????
- (you have a fixed symbol)
- one symbol per empty grid space
- don't erase symbols

O	O	O
	O	X
	X	X

Josef Who?

What are game rules?

Example: Tic Tac Toe

These simple rules describe the formal system of the game

They don't describe the experience of playing the game

They don't describe the history and culture of the game

They just constitute the rules of the game

**Rules are the deep structure of a game
from which all real-world instances of the
game's play are derived**

What are game rules?

What exactly are and are not rules?

Josef Who?

What are game rules?

Example: Poker

What exactly are and are not rules?

Yes:

- how many cards are dealt to the player
- how to bet
- value of card combinations

Card deck, yes:

- exact number of cards
- card defined by number and suit



Josef Who?

What are game rules?

Example: Poker

What exactly are and are not rules?

Yes:

- how many cards are dealt to the player
- how to bet
- value of card combinations

Card deck, yes:

- exact number of cards
- card defined by number and suit
- value of card combinations

Card deck, no:

- 4 suits: Spades, Hearts, Clubs, Diamonds



What are game rules?

Rules and Strategy

We distinguish between:

- rules of a game -> formal structure**
- rules of strategy -> strategies for play**

Rules that help players perform better are not part of the formal system.

What are game rules?

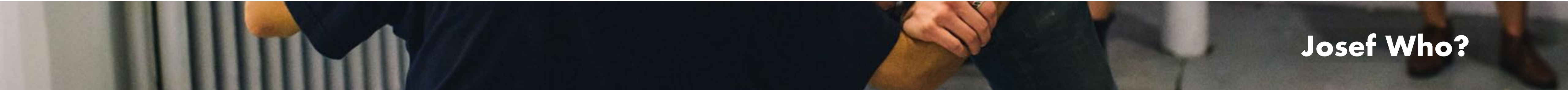
What makes game rules unique?

Games are artificial systems, separate from ordinary life

Magic Circle

Marc Prensky:

“Rules are what differentiate games from other kinds of play. Probably the most basic definition of a game is that it is organized play, that is to say rule-based. If you don't have rules you have free play, not a game. ”



Josef Who?

How do game rules function?

Game Rule Characteristics:

- Rules limit player action



Josef Who?

How do game rules function?

Game Rule Characteristics:

- Rules are explicit and unambiguous



Josef Who?

Game Rule Characteristics:

- Rules are shared by all players
also their interpretation!

How do game rules function?



Josef Who?

How do game rules function?

Game Rule Characteristics:

- Rules are fixed



Josef Who?

How do game rules function?

Game Rule Characteristics:

- Rules are binding

otherwise they wouldn't be fixed and shared



Josef Who?

How do game rules function?

Game Rule Characteristics:

- Rules are repeatable (and portable)
from game to game and from player set to player set



Josef Who?

How do game rules function?

Game Rules

- limit player action
- are explicit and unambiguous
- are shared by all players
- are fixed
- are binding
- are repeatable

If any of these characteristics are not in effect, the game system may break down, making play impossible.

Rules on 3 levels

Are there rules outside of the formal rules?

Do the formal rules constitute the complete formal system?

0	0	0
	0	X
	X	X

Rules on 3 levels

There are Under the Hood Rules

and

being a good sport rules.

Rules on 3 levels

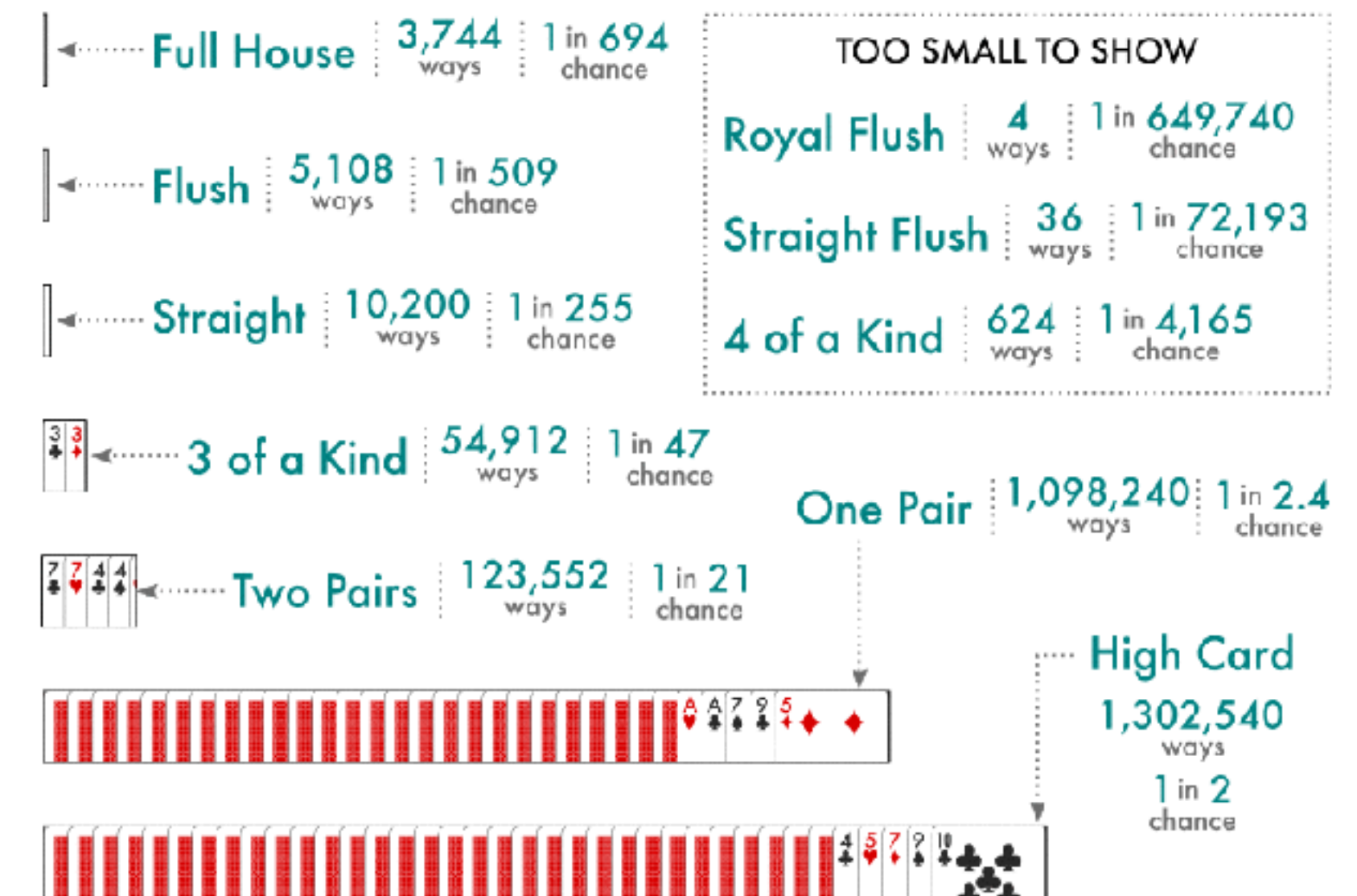
Under the Hood Rules

Systems feeding the stated rules



POKER HAND PROBABILITIES

There are **2,598,960** unique five-card hands in a standard 52-card deck.
This graph shows the number of different ways each poker hand can be formed and lists the probability of each being dealt as the first five cards.



Source: Scarne on Cards by John Scarne

www.denexa.com

NO LIMITS

Denexa™
GAMES

Josef Who?

Rules on 3 levels

Being a Good Sport Rules

Lie beyond or outside stated rules

Oftentimes behavioural rules



Josef Who?

Rules on 3 levels

- Operational Rules
- Constitutive Rules (under the hood/below the surface)
- Implicit Rules (behavioural/being a good sport)

Rules on 3 levels

Fingerübung 🖐️

- Operational Rules
- Constitutive Rules
- Implicit Rules



Josef Who?

Rules on 3 levels

Fingerübung 🤞

- Operational Rules (tangible)
- Constitutive Rules (abstract)
- Implicit Rules (unexpressed)

Operational and constitutive rules together form the true and unique identity of the game whereas the implicit ones don't

BUT also does the material!

What is the game rules' relevance for game design?

You generally want players focused on the experience of play, rather than on making sense of the rules.

Rules are one of the raw materials of game designers.

The more you know about rules the more elegant and sharp your design can be to achieve and maintain proper player focus.

Rules

Josef Who?



Interactivity

Josef Who?



Do you know the game?



Genital Jousting

Free Lives

Play implies interactivity:

**To play with a game, a toy, a person, an idea,
is to interact with it.**

More specific:

Playing a game means making choices within a game system designed to support actions and outcomes in meaningful ways.

This action and outcome feedback loop is interactivity, with the designed system of the game.

Interaction happens on many levels, e.g.

- with game's objects/pieces/content**
- social interaction**
- cultural interaction**

What is interactivity?

Which aspects of interactivity are relevant for Game Design?

Interactivity describes an active relationship between two things.

Characteristics of interactivity in games:

- players perform **explicit actions**
- players make **choices** that have been designed into the structure of the game
- with those actions and choices they start a **conversation** with the game's system
- takes place within a **system**
- is iterative

Designing interaction

2 action examples:

- someone dropping an apple
- someone rolling a dice

Questions:

Which one is a designed interaction?

What about this action has been designed?



Designed Interactivity

What conditions/context would have to be present to evolve a simple interaction into a designed interaction?

Design creates meaning!



Quality of interaction

The relationship between the player's choice and the system's response is one way to characterize the depth and quality of interaction.

Choices can be designed on two levels: micro (moment-to-moment) and macro (larger trajectory)



Choices

Every choice has an outcome.

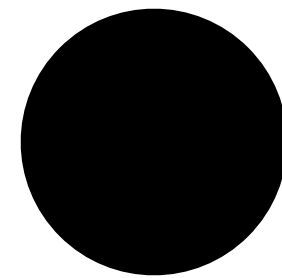
Choice follows an action $\rightarrow$ outcome unit/molecule

A game is a state machine that is influenced by players

There can be an internal and external state

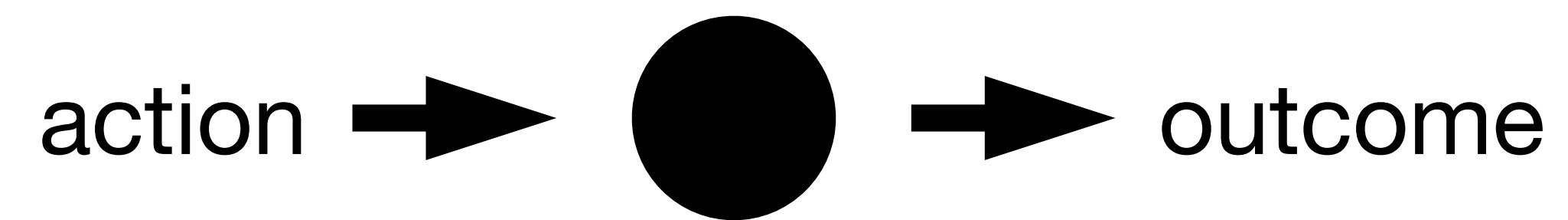
System

Interactivity



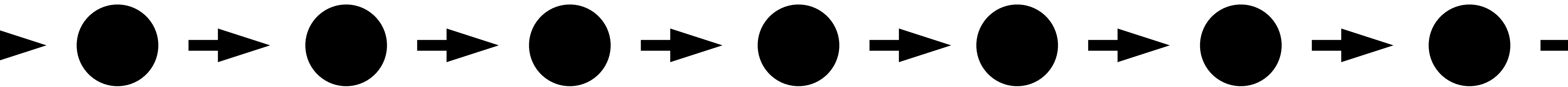
Josef Who?

Choice



Interactivity

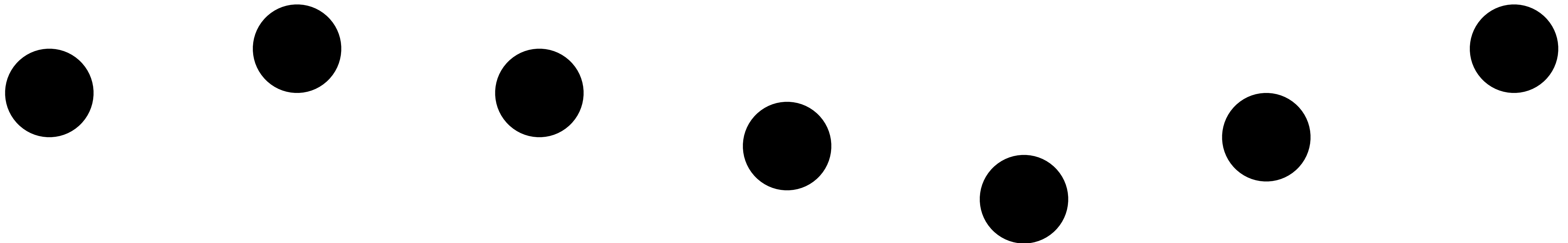
Choices



Josef Who?

Choices chain

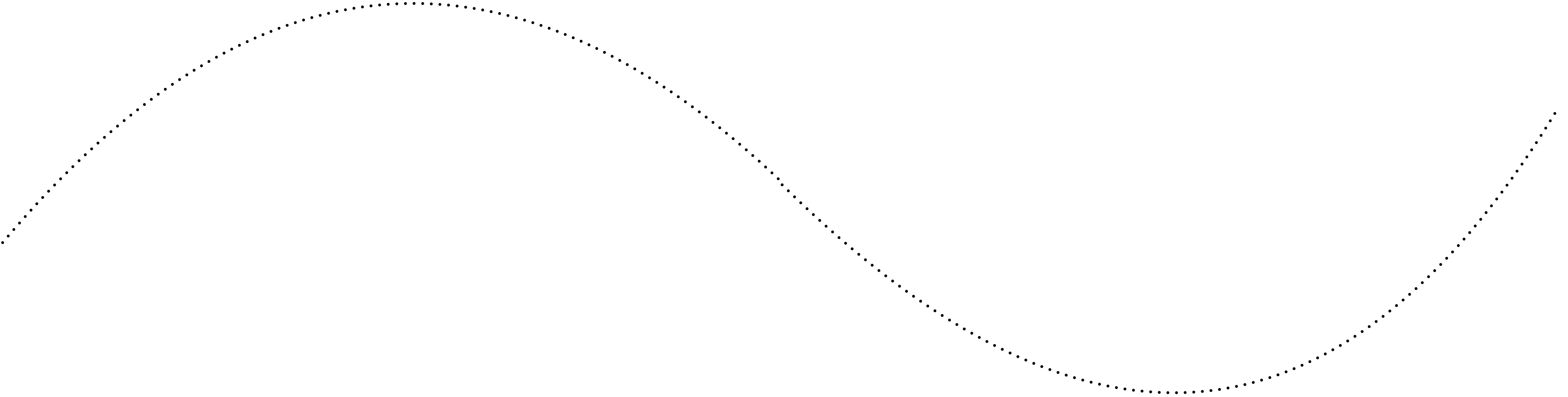
Interactivity



Josef Who?

Interactivity

Macro Level



Josef Who?

Space of possibility

Game designers do not directly design play

They design structures and context

They indirectly shape the actions of players



Interactivity

Josef Who?



Hands-On Elemental Duel

Josef Who?



Do you know the game?



SIFU

Slocap

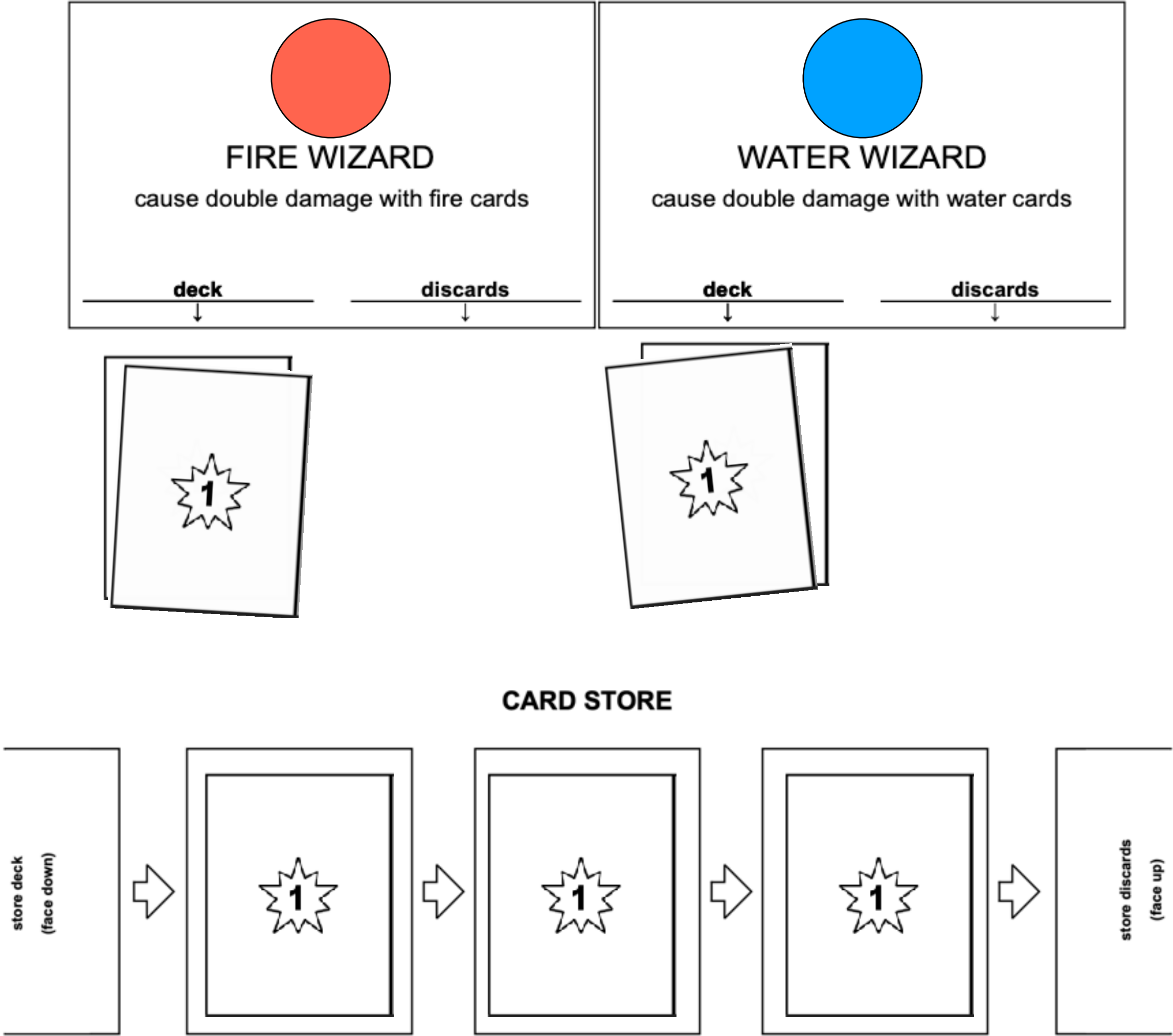
Elemental Duel, Eric Zimmerman

Iterative process

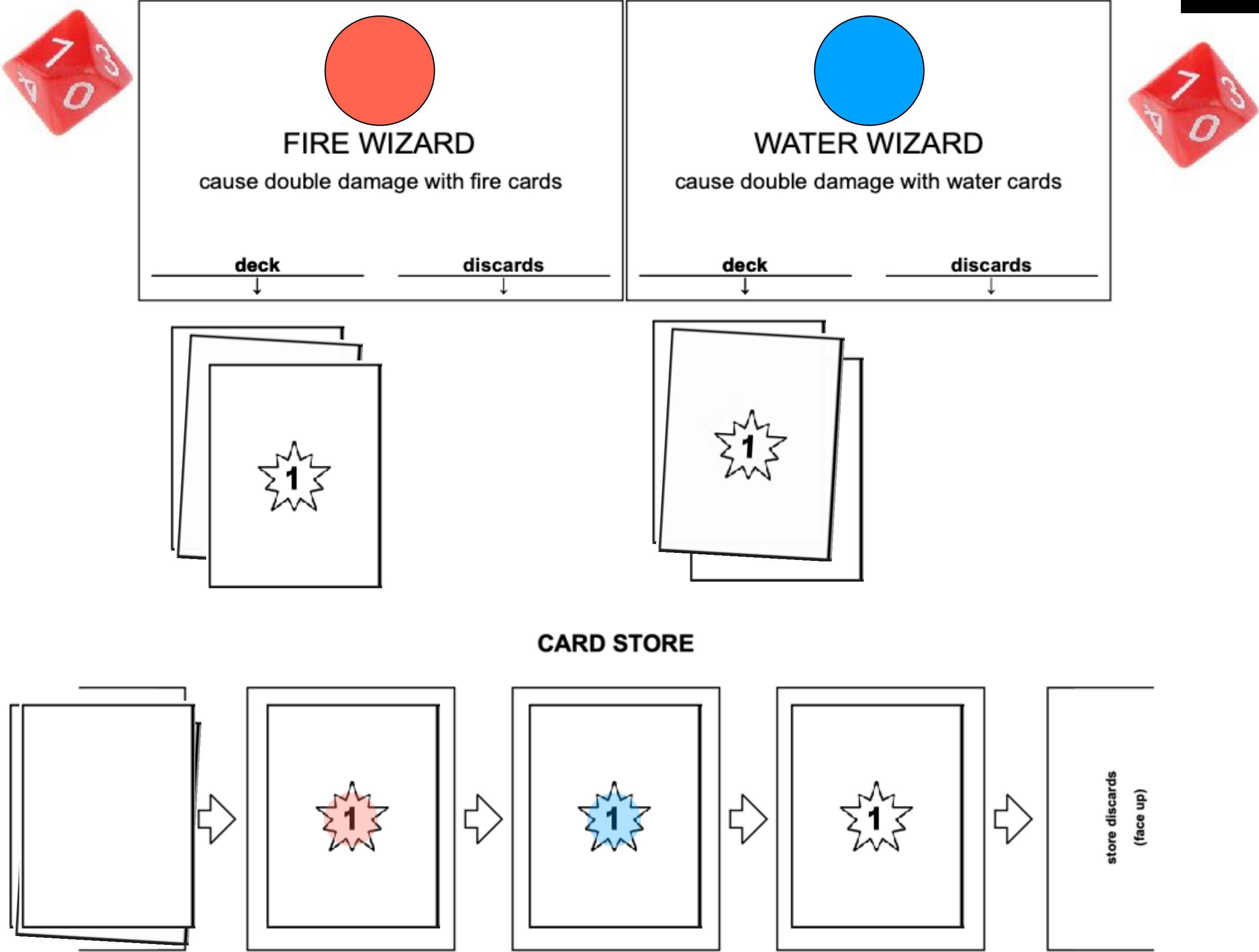
- play
- discuss
- mod
- playtest
- present

Elemental Duel

Premise and Material



Elemental Duel
Setup



Elemental Duel

Water and Fire

CARD EFFECTS



BASIC

opponent takes immediate damage



FIRE

1. add burn tokens to opponent = card #
2. at the start of your next turn:
they take damage = total burn tokens
3. then remove 1 token from opponent



WATER

1. opponent takes immediate damage
2. at the end of your turn, heal health equal to
one less than the total of any water damage
caused this turn

Elemental Duel

Steps per Round

ON YOUR TURN:

1. Apply fire burn damage.
2. Draw up to a hand of 2 cards (on the 1st player's 1st turn draw 1).
3. Play any or all of your cards, discarding cards played.
4. If you need to draw and your deck has run out, reshuffle discards into your deck.
5. Select any one card from the store and place it in your discards.
6. Heal health from water cards played this turn.

Elemental Duel

End of Round/Card Store

AT THE END OF YOUR TURN:

Move all cards in the store one space along the arrows.

Any card that moved off the store is put in the store discard.

Place new cards in the store to fill it up.

If the store deck runs out, reshuffle the discards and put it back on the deck.

Elemental Duel

Play one round and then we discuss the game

You have time until ???

Observations Gruppe A

fun but not enough HP; too short; not enough strategy therefore; person who starts wins ???; special cards useless to other player; beginning very stale, first turns always the same; no reason to not play all cards; all in all unbalanced; elementals either overpowered or no meaning at all; fire for many overpowered;

Observations Gruppe B

needs more work; beginner has a disadvantage ?; little choice; you should use your own element; slow start, boring start; makes no sense to not play all of your cards; card probabilities maybe off?; fire has an advantage?; water has an advantage?; not enough choices; good for kids; only choice is what do I pick from the store; blue cards in hand of fire wizard not that interesting;

Elemental Duel

Mod the game in groups of 3

You can:

- change the rules
- change setup
- change sequences
- add/subtract rules/roles,...
- add/subtract material

Elemental Duel

Mod the game in groups of 3

Do not:

- change everything but also keep stuff
- change too much
- think all your changes through to the last bit, instead try them out immediately
- hold onto ideas that don't work -> kill your darlings
- underestimate time

Elemental Duel

Mod the game in groups of 3

You have to:

- write down your rules
- produce all materials needed for the game

Playtest

send teammates away

one has to stay behind

let the newbies play without saying anything

just watch, don't talk!

your instructions and materials have to do the talking.

-> design is also communication!

Elemental Duel

Mod the game in groups of 3

You can:

- change the rules
- change setup
- change sequences
- add/subtract rules/roles,...
- add/subtract material

Let's go!

You have time until ???

Playtest

send teammates away

one has to stay behind

let the newbies play without saying anything

just watch, don't talk!

your instructions and materials have to do the talking.

-> design is also communication!

Discuss changes

What did you change?

What did you keep?

What did you add?

How was your process?

What was the feedback?

Did the testers understand everything?

What worked and what didn't work?

Results Gruppe A

- 1)
8 cards in the beginning face down; and then everybody chooses 4;
health now 30; 3 cards max in hand, max draw 2; trifactor special effect:
all cards at the same time double effect;
- 2)
one shared deck and discard pile; removed the shop completely; dice
throw to determine starting player; limit to up to 2 cards played; basic 1
have different effects (healing, reshuffling, ...); star symbol cards adds
token additionally to the other token; no double damage; fire and water
cards only effect with corresponding class;

Results Gruppe A

3)

start with 1 basic and 2 corresponding cards; health up to 20; very first plays no cards; every round 2 cards drawn; shop refreshes every turn; basic card is a re-roll card now; re-roll shop or get one extra shop card; water heals not with -1 anymore; negate card added; max hand size is 7; cards from store directly in hand;

4) no wizards, both the same; 2 basic 1 and one of each element at the start; 20HP; both take 2 cards in the beginning; you can play cards or get from shop or mix; changed card effects a lot; only 1 of each element each round; reworked fire: burns a card of the opponent; added lightning which reduces the action of the opponent;

Results Gruppe A

5)

classes gone; bumped numbers, no basic 1; draw up to 3 and play up to 2; 2 of the same colour you get the effect; start with 1 basic, 1 water, 1 fire;

6) health 20; hold up to 5 and play as many as you like; shop moves 2 to the right -> refreshes more often; 2 fuck you cards: negate cards; you still lose 1 health with fuck you;

Results Gruppe B

1)

twice HP; one basic, one fire, one water for the start; max of 3 in hand, draw 2, play max 3; reshuffle discard only when you start with 0 cards in deck; added 3 special cards: block, thief, collect; they get back into store discard; shop is not fully refilled;

2)

no additions; start with some deck building; start with 5 cards; 3 cards in hand max, only play 2 max; store changes: look at 3, keep one, put one in store and put one in store discard; fire and water effects: fire places 2 tokens, water cards can remove fire tokens; water still heals;

Results Gruppe B

3)

store phase: pick 1 and also swap one from your deck with one from store; more deckbuilding; you can not play cards and cards go back in deck so you can swap them later; store cards go directly into your deck; HP 20 now; 2 decks combined;

4)

same setup; draw 2 but up to max 3 in hand; play only 1; additionally exchange one from hand with one from store; max 2 damage per round; 6 new card types with special effects;

Results Gruppe B

5)

re-did store; leftover cards in facedown pile that are now the new store; 3 on top are the ones to have access to; start with 4 cards (3 basic one elemental); reshuffling means you can get rid of one bad card -> to store;

6)

added shield; every damage gives you some kind of shield; tokens for shield; only for basic attacks; no discard in store; just replaced with a card from store deck; at start: 1 water, 1 fire, 1 basic; 3 cards on hand; burn damage at end of your round;

The background is a vibrant, low-poly 3D game environment. A large, glowing red sun dominates the upper left, casting a warm, reddish light over the scene. The terrain is composed of various low-poly shapes, including rocky outcrops and a path. A character, possibly a skateboarder, is visible in the lower right, moving along a path. The overall aesthetic is reminiscent of early 2000s indie games.

Game Studies & Game Design

Josef Who?